

AURA

Autonomous University Response and Action

ACADEMIC PRESENTATION

Advanced Programming Java Project

Department of Computer Science & Engineering • TKM College of Engineering (TKMCE)

AURA

Autonomous University Response and Action

"REPORT. TRACK. RESOLVE. IMPROVE."

PROJECT TEAM MEMBERS & SLIDE ASSIGNMENTS

Advanced Programming Java Project

B25CSB45

LEAD

MUHAMMED RINSHID VP

Team Lead

Assigned: Slides 1-4, 20

B25CSB52

NIRMAL BINOY

Core Developer

Assigned: Slides 5-8

B25CSB37

MOHAMMED NAFIH

Core Developer

Assigned: Slides 9-12

B25CSB53

RAHANDEEP RD

Database & GUI Engineer

Assigned: Slides 13-14, 18

B25CSB84

ATHIL RAHUMAN A

Documentation & Process Lead

Assigned: Slides 15-17, 19

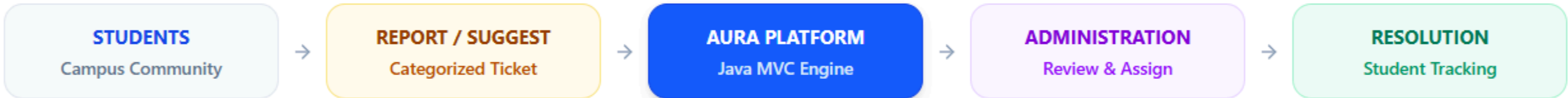
What is AURA?

Project Overview & Core Mission

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AURA END-TO-END OPERATIONAL LIFECYCLE



● Centralized Governance Portal

Consolidates student issues, infrastructural complaints, and institutional suggestions into one unified Java platform.

● Role-Based Access Control

Strict separation of privileges between Student submission/tracking interfaces and Admin resolution dashboards.

● Transparent Workflow Pipeline

Live state updates from PENDING to ASSIGNED, IN_PROGRESS, and RESOLVED with admin resolution notes.

● Pure Java MVC Architecture

Built strictly with standard Java SE, Swing/JavaFX Desktop GUI, JDBC DAO patterns, and MySQL relational persistence.

Motivation & Problem Statement

The Need for Structured Campus Governance

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⚠ Current Campus Challenges

FRAGMENTED

- ✗ **Informal Channels:** WhatsApp messages and word-of-mouth complaints get buried or forgotten.
- ✗ **Zero Progress Tracking:** Students have no visibility into issue status after submission.
- ✗ **Lack of Accountability:** No clear assignment of administrative staff or completion metrics.
- ✗ **No Analytics:** Administration lacks aggregated data on recurring infrastructural bottlenecks.

✓ AURA Transformation

STRUCTURED

- ✓ **Centralized System:** Single platform for all student issues and campus suggestions.
- ✓ **Real-time Status Pipeline:** Automated updates from submission through final resolution.
- ✓ **Role-Based Ownership:** Clear staff assignment and transparent resolution logging.
- ✓ **Data-Driven Reports:** Administrative dashboards for proactive campus maintenance.

Sustainable Development Goals Alignment

How AURA contributes to the United Nations Sustainable Development Goals



 **AURA transforms student feedback into structured institutional action — supporting a more responsive, inclusive and well-managed campus.**



Existing Systems & The Gap

Evaluating Current Channels and Identifying Bottlenecks

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COMPARISON MATRIX: EXISTING METHODS VS AURA PLATFORM				
Channel	Centralized?	Live Tracking?	Role Security?	Primary Limitation
WhatsApp / Messaging	No	No	No	Unstructured, messages lost in group chats
Paper Suggestion Boxes	Partial	No	No	Delayed collection, zero feedback to student
Google Forms	Yes	No	Basic	Flat sheet, lacks resolution lifecycle states
✔ AURA Platform	Yes (Java SE)	Yes (5-State)	Yes (RBAC)	Purpose-built end-to-end campus governance

PROJECT UPGRADE PRESENTATION • SLIDE 05

Proposed Solution

End-to-End Governance Architecture

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1

Secure Authentication

Role-based login for Students and Administrators with password validation and session management.

2

Issue & Suggestion Entry

Categorized ticket submission specifying issue type, priority level, title, and detailed description.

3

Live Status Engine

State machine transitioning tickets through PENDING, ASSIGNED, IN_PROGRESS, RESOLVED, or REJECTED.

4

Admin Control Panel

Staff assignment controls, status update hooks, resolution note recording, and submission filtering.

5

DAO & Relational Store

Clean Data Access Object layer utilizing JDBC connections for normalized MySQL persistence.

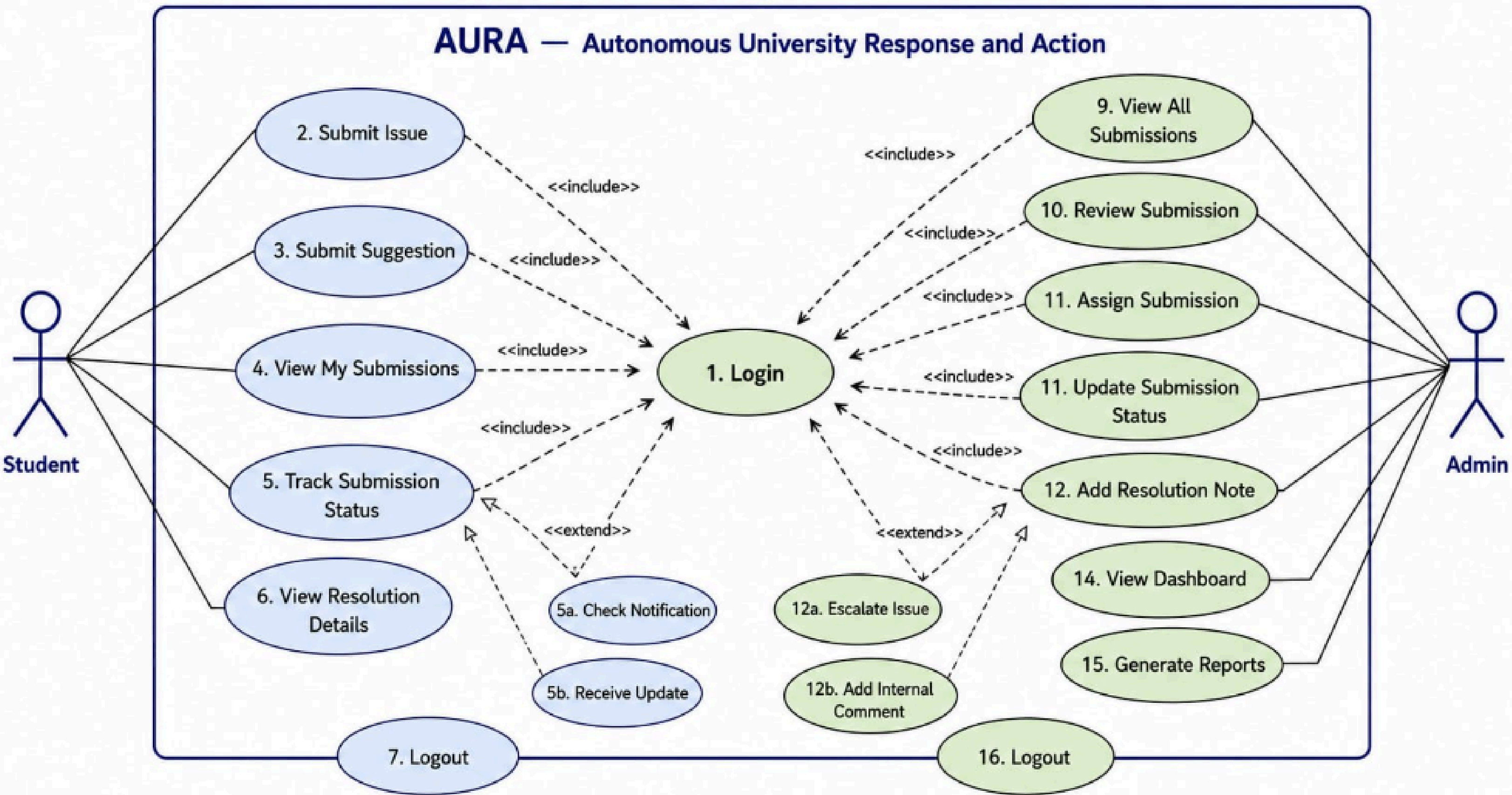
6

Desktop GUI Interface

Responsive Java Swing/JavaFX user interface optimized for high productivity on desktop environments.

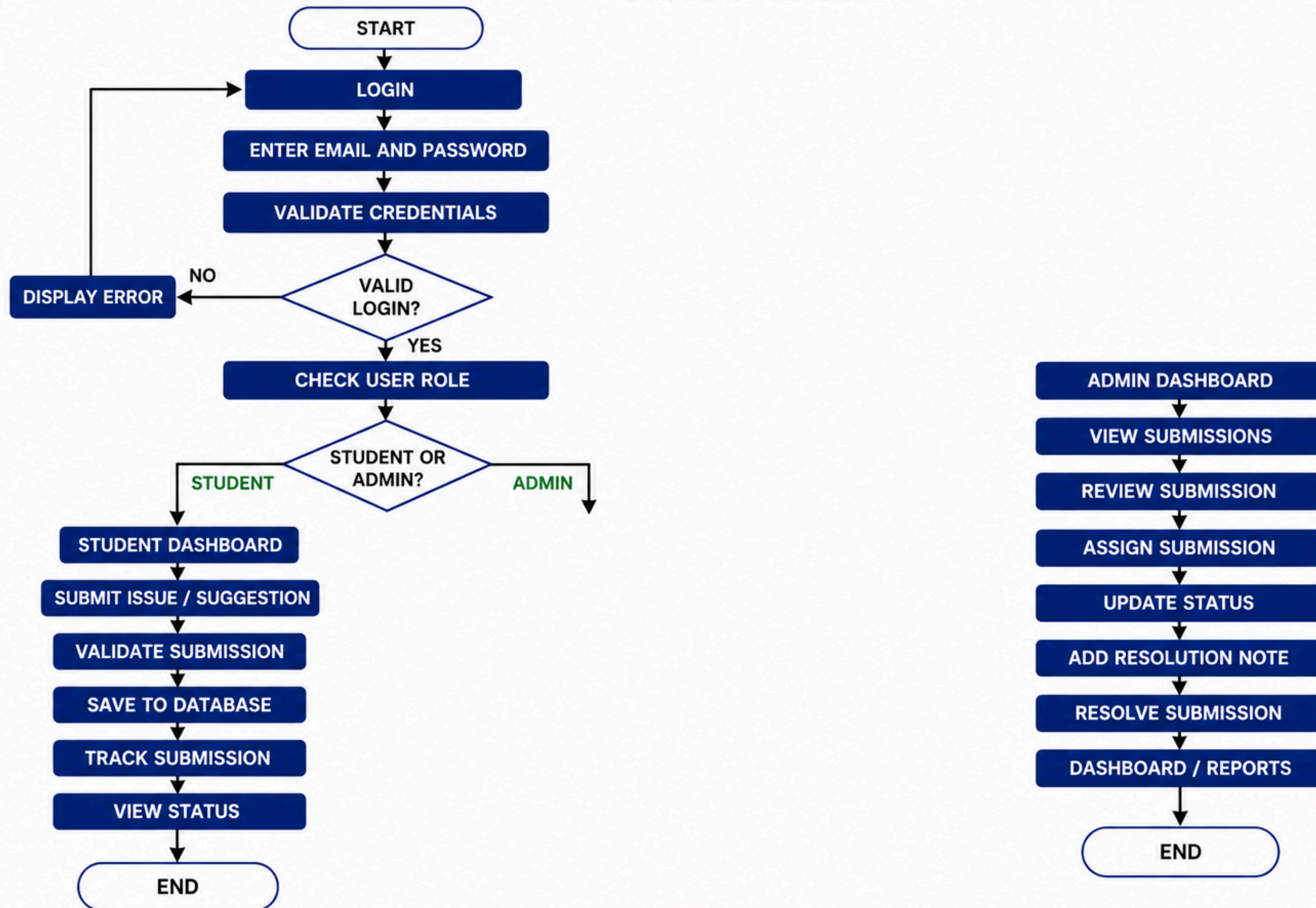
UML Use Case Diagram

System Actors and Functions

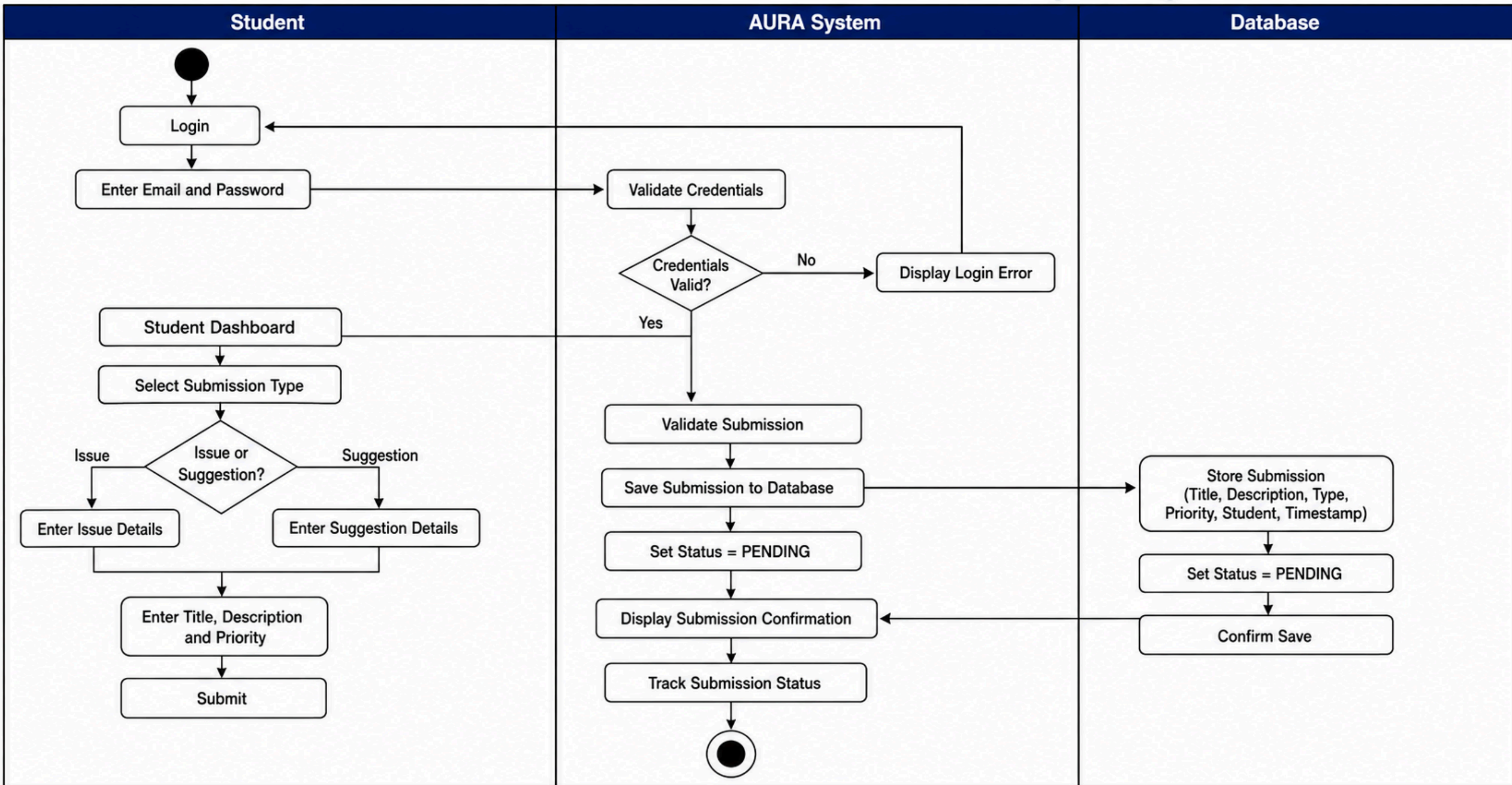


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SYSTEM FLOWCHART

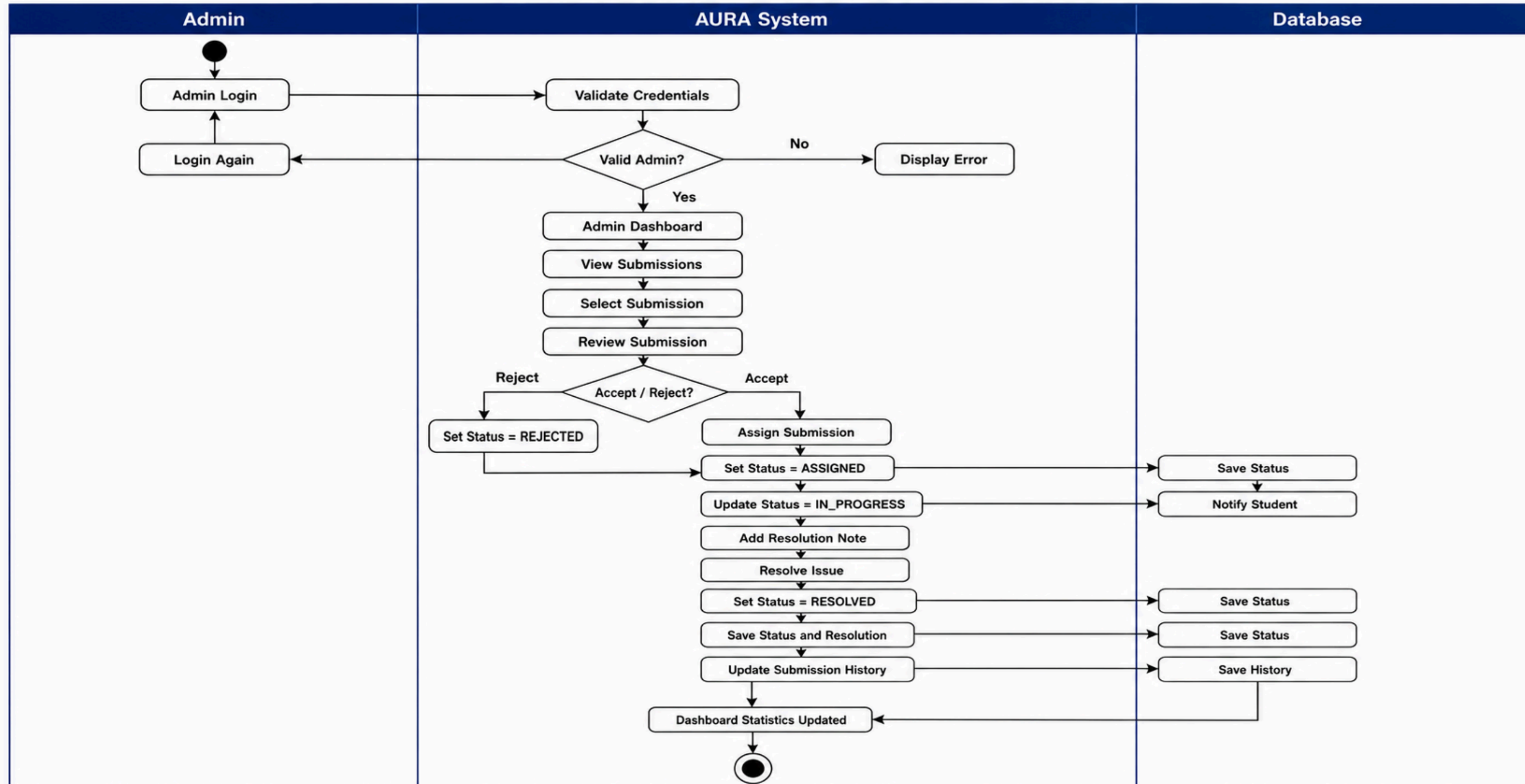


AURA – Student Submission Activity Diagram



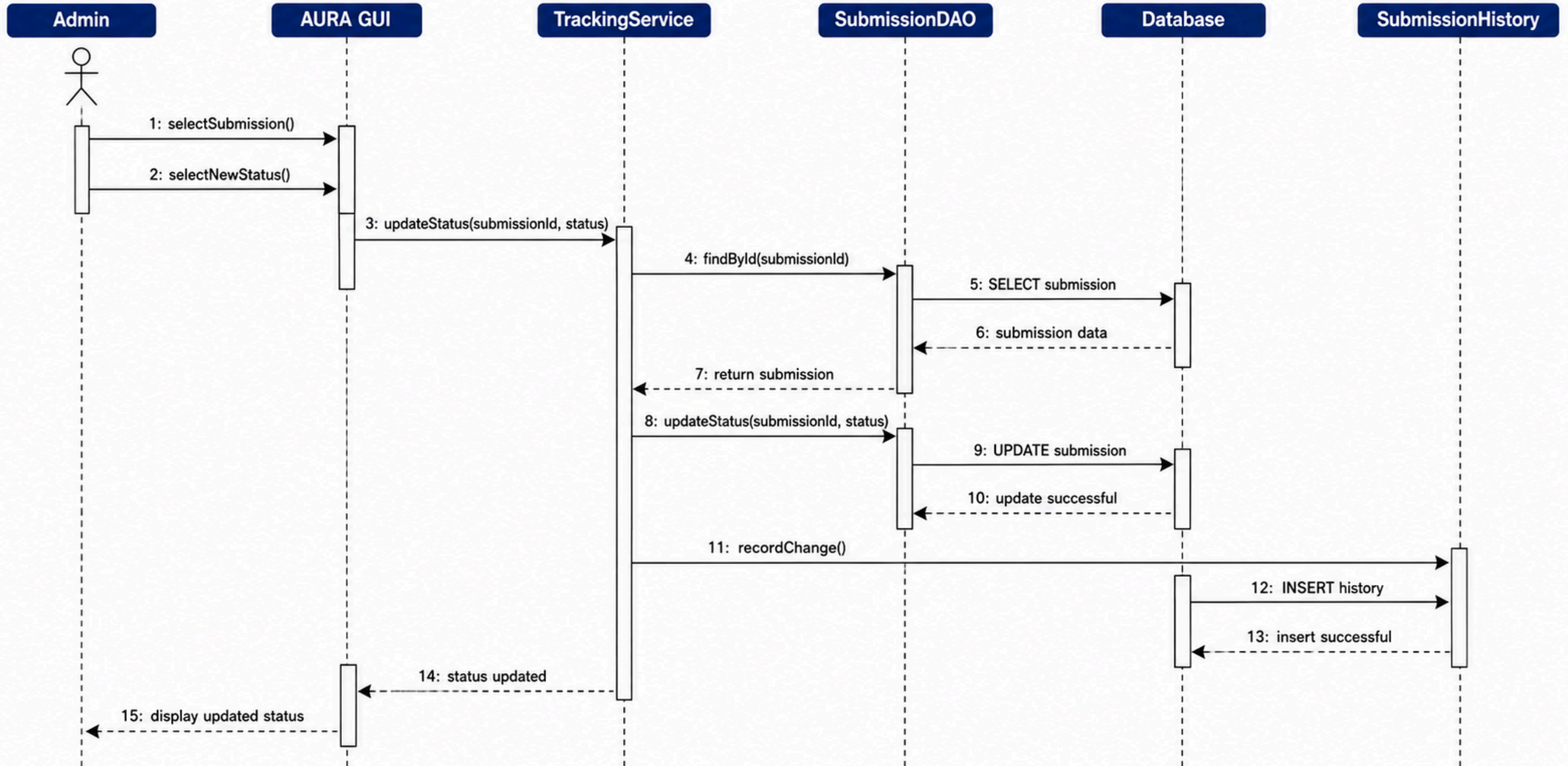
AURA — Admin Submission Management Activity Diagram

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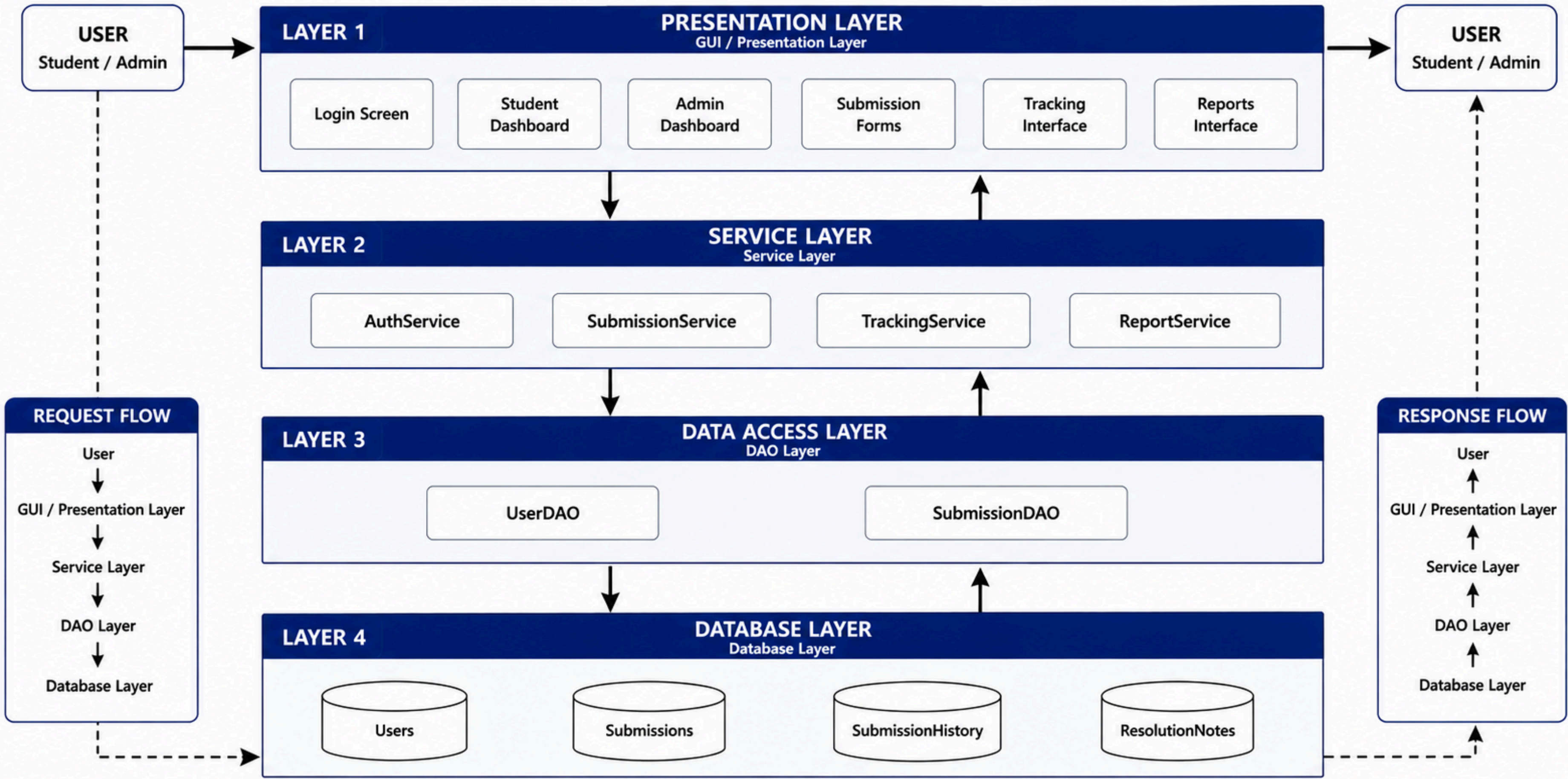
AURA — UML SEQUENCE DIAGRAM

Admin Status Update and History Tracking



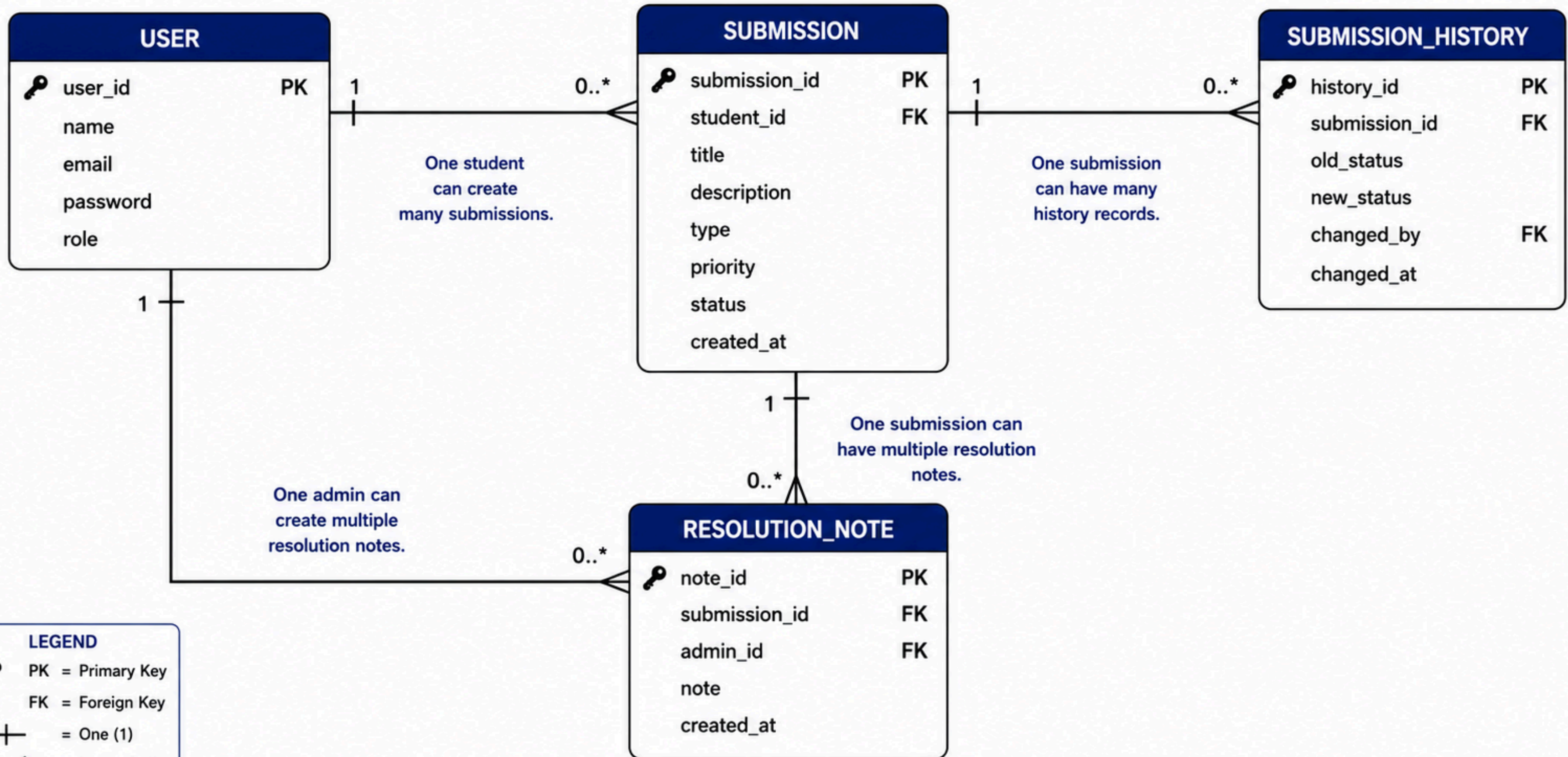
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Layered System Architecture



AURA — DATABASE / ER DIAGRAM

AURA — Autonomous University Response and Action



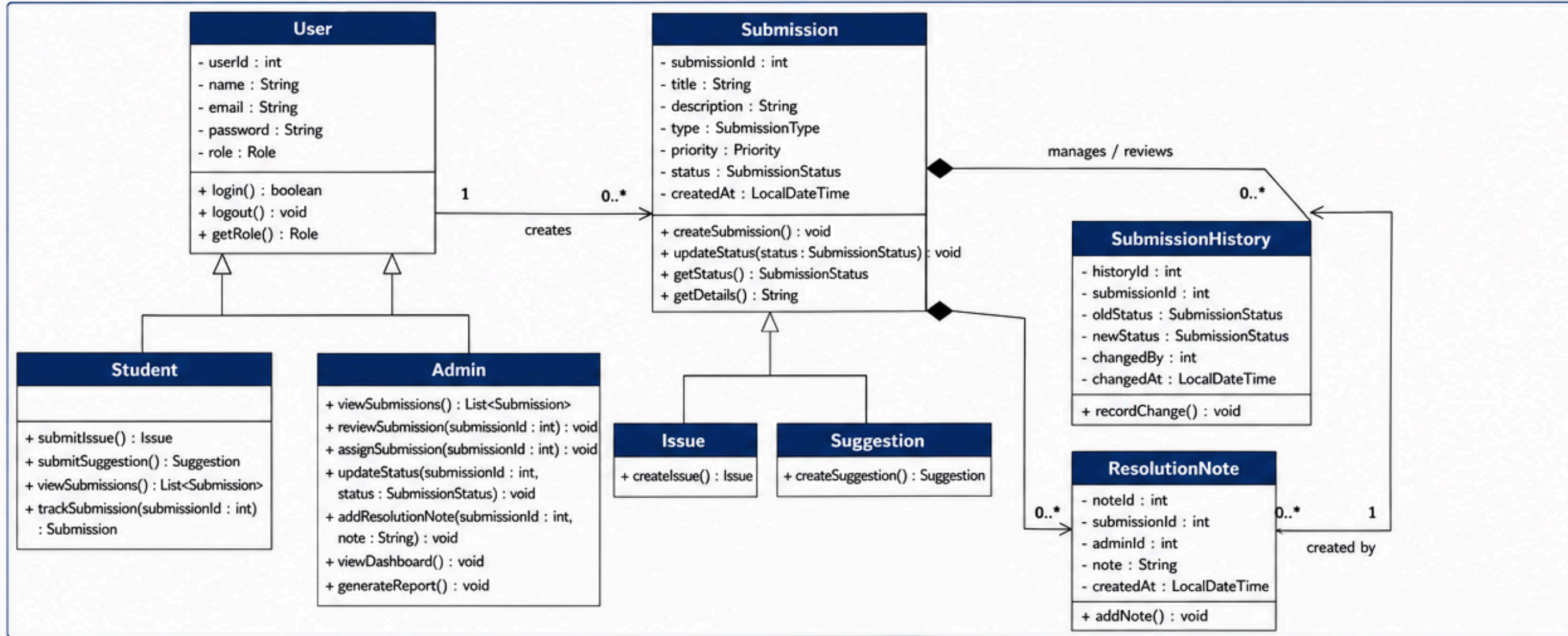
LEGEND

- PK = Primary Key
- FK = Foreign Key
- + = One (1)
- Y = Many (0..*)

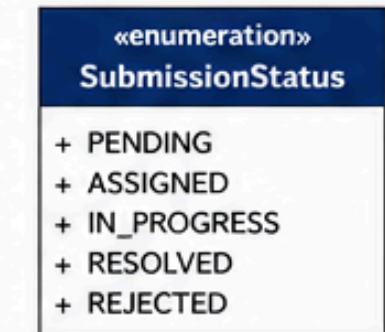
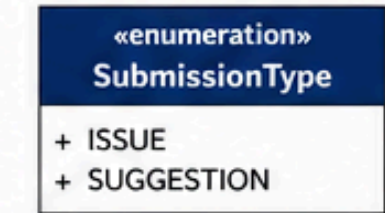
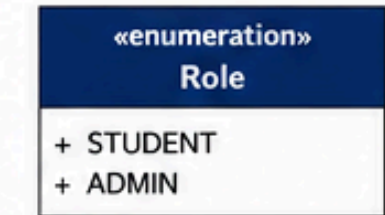
UML CLASS DIAGRAM

Java Advanced Programming Project

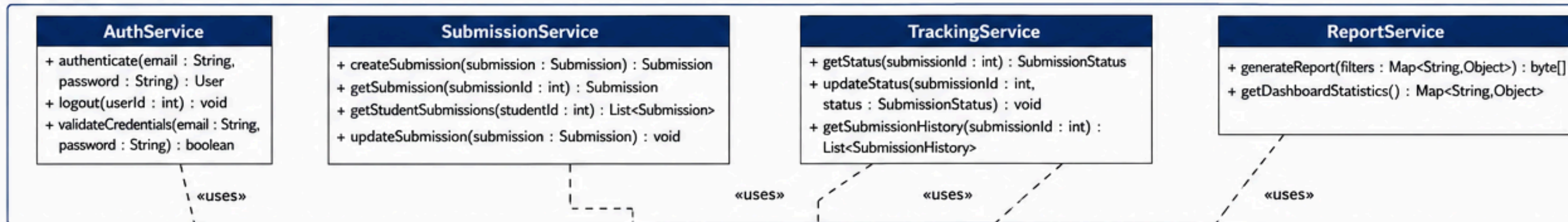
«package» model



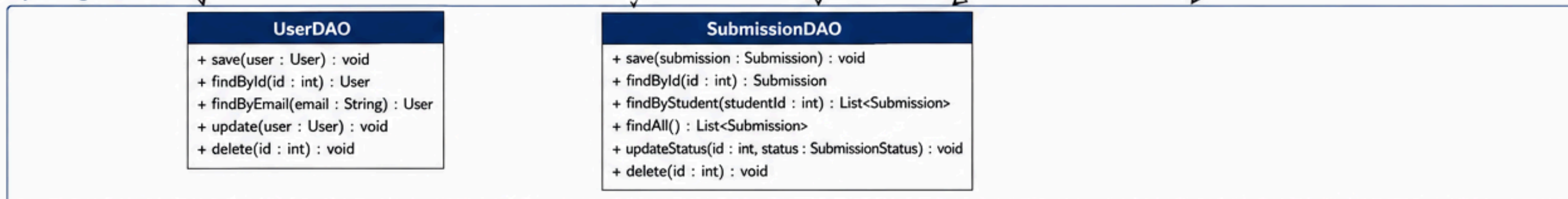
«package» enum



«package» service



«package» dao

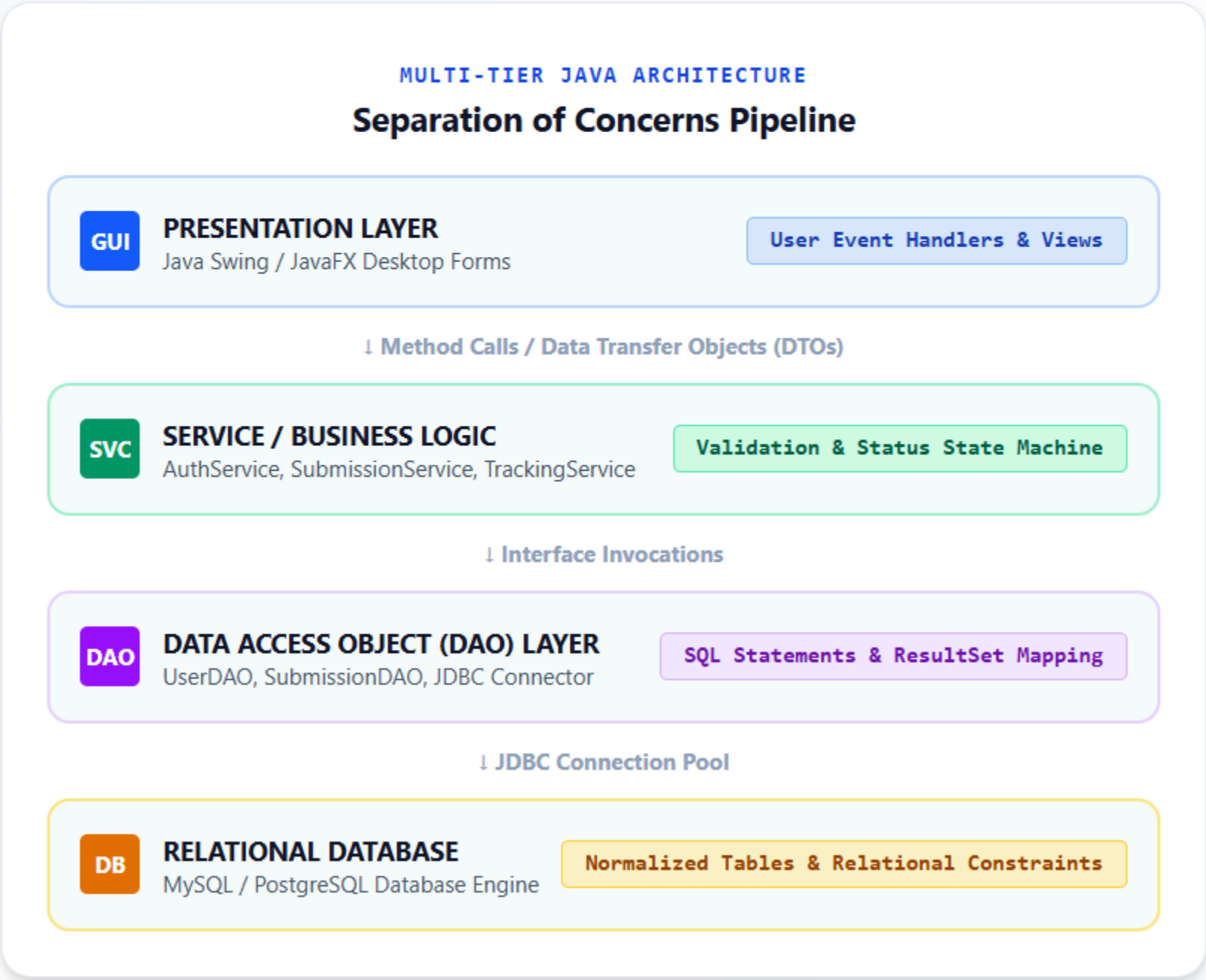


System Architecture

Clean Layered Software Design (GUI → Service → DAO → Database)

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User / Role Architecture

Role-Based Access Control (RBAC)

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STUDENT ROLE

`Class Student extends User`

Permitted Actions:

- ✓ `Submit Campus Issues`
- ✓ `Submit Institutional Suggestions`
- ✓ `Track Own Submission History`
- ✓ `View Resolution Notes`



ADMINISTRATOR ROLE

`Class Admin extends User`

Permitted Actions:

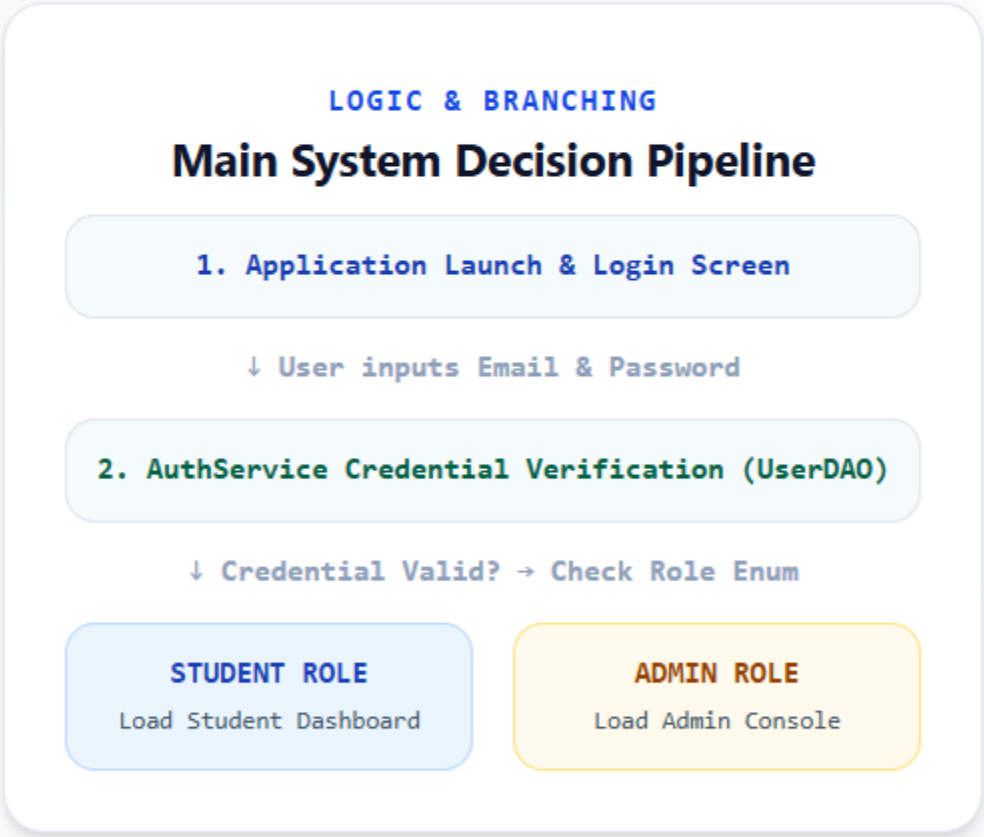
- ✓ `Review All Campus Submissions`
- ✓ `Assign Staff / Department Handlers`
- ✓ `Update Status (IN_PROGRESS, RESOLVED)`
- ✓ `Attach Official Resolution Notes`
- ✓ `Generate System Analytics & Reports`

Main System Flowchart

High-Level Logic & Decision Pipeline

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Student Workflow

Issue & Suggestion Lifecycle

STUDENT EXPERIENCE

Issue & Suggestion Lifecycle



1. Access Dashboard

Student authenticates into the Java desktop application and views active submissions.



2. Fill Submission Form

Selects Type (ISSUE / SUGGESTION), Priority (LOW / MEDIUM / HIGH), Title, and Description.



3. DAO Submission

SubmissionDAO saves ticket into MySQL with initial status set to PENDING.



4. Live Tracking

Student monitors status changes and reads official resolution notes upon completion.



Admin Workflow

Review, Assignment, and Resolution Execution

ADMINISTRATIVE PROCESSING

Review, Assignment, and Resolution Execution

1. Admin Management Console

Administrator reviews incoming tickets sorted by Priority and Submission Date.

2. Staff Assignment

Assigns responsible staff department and updates ticket status to ASSIGNED / IN_PROGRESS.

3. Attach Resolution Note

Upon completion, writes resolution details and sets status to RESOLVED.

4. Dashboard Refresh

System analytics automatically recalculate resolution metrics and display overall university progress.

GUI Design

Java Desktop Application Interface Mockups

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1. Login Screen

2. Student Dashboard

3. Submit Issue / Suggestion

4. Admin Dashboard

AURA Desktop Client v1.0.0 — JavaFX Applet [JFrame]

FileEditViewToolsHelp

Connected: localhost:3306/aura_db

A

AURA — User Login

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Role / Profile:

STUDENT — Student Access

Campus Email Address:

student@university.edu

Password:

.....

Authenticate & Launch Dashboard →

JVM Engine Ready | MVC Active | JDBC Pool: Connected

Swing / JavaFX LookAndFeel: Nimbus / Metal

GUI Design

Java Desktop Application Interface Mockups

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1. Login Screen
2. Student Dashboard
3. Submit Issue / Suggestion
4. Admin Dashboard

J AURA Desktop Client v1.0.0 — JavaFX Applet [JFrame]

File

Edit

View

Tools

Help

Connected: localhost:3306/aura_db

Logon Session: Student #20260481

+ Submit New Ticket

Student Issue & Suggestion Management Console

TOTAL SUBMISSIONS

4

PENDING

1

IN PROGRESS

2

RESOLVED

1

JTable: Recent Submissions Record

Query Result: 4 records

ID	TYPE	TITLE	PRIORITY	STATUS	DATE
#SUB-104	ISSUE	Lab 3 Projector Lamp Faulty	HIGH	IN_PROGRESS	2026-08-07
#SUB-103	SUGGESTION	Extend Library Hours for Midterms	MEDIUM	PENDING	2026-08-06
#SUB-098	ISSUE	Wi-Fi Disconnections in Block B	HIGH	RESOLVED	2026-08-04
#SUB-089	SUGGESTION	Add Water Coolers on 2nd Floor	LOW	ASSIGNED	2026-08-01

JVM Engine Ready | MVC Active | JDBC Pool: Connected

Swing / JavaFX LookAndFeel: Nimbus / Metal



GUI Design

Java Desktop Application Interface Mockups

1. Login Screen

2. Student Dashboard

3. Submit Issue / Suggestion

4. Admin Dashboard

AURA Desktop Client v1.0.0 — JavaFX Applet [JFrame]

FileEditViewToolsHelp

Connected: localhost:3306/aura_db

Create Submission (JPanel Form)

AURA.SubmissionService

Submission Type:

ISSUE

SUGGESTION

Priority Level:

HIGH

Title / Brief Summary:

e.g., Air Conditioner leaking in Seminar Hall 2

Detailed Description:

Provide specific details, room numbers, and relevant info...

Cancel

Submit to DAO / JDBC

JVM Engine Ready | MVC Active | JDBC Pool: Connected

Swing / JavaFX LookAndFeel: Nimbus / Metal

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GUI Design

Java Desktop Application Interface Mockups

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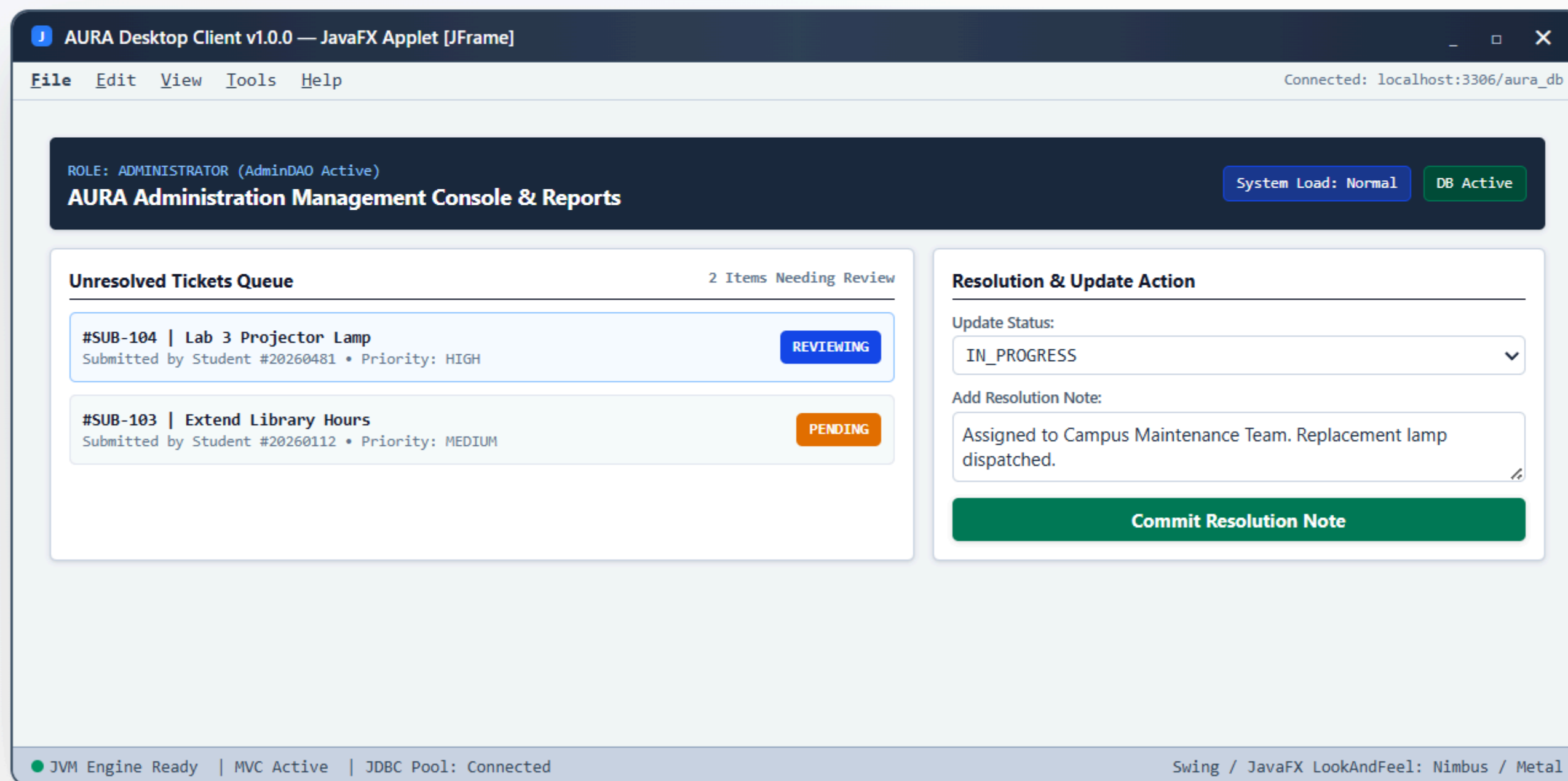
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1. Login Screen

2. Student Dashboard

3. Submit Issue / Suggestion

4. Admin Dashboard



Database / ER Diagram

Relational Schema and Entity Mapping

RELATIONAL PERSISTENCE SCHEMA

Entity Relationship (ER) Mapping

TABLE: users [PK] user_id name email password role INT VARCHAR(100) VARCHAR(150) VARCHAR(255) ENUM('STUDENT', 'ADMIN')	TABLE: submissions [PK] submission_id [FK] student_id title description type priority status created_at INT INT VARCHAR(200) TEXT ENUM('ISSUE', 'SUGGESTION') ENUM('LOW', 'MEDIUM', 'HIGH') ENUM('PENDING', 'ASSIGNED', 'IN_PROGRESS', 'RESOLVED', 'REJECTED') TIMESTAMP
TABLE: submission_history [PK] history_id [FK] submission_id old_status new_status [FK] changed_by changed_at INT INT VARCHAR(50) VARCHAR(50) INT TIMESTAMP	TABLE: resolution_notes [PK] note_id [FK] submission_id [FK] admin_id note created_at INT INT INT TEXT TIMESTAMP



Development Progress

Current Status and Verification Matrix

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COMPLETED DELIVERABLES

- ✓ Project idea & scope finalized
- ✓ Official AURA branding set
- ✓ Product Requirements (prd.md)
- ✓ Technical Requirements (trd.md)
- ✓ System Architecture (architecture.md)
- ✓ GitHub Repository initialized
- ✓ Complete UML Class & Use Case Diagrams



PLANNED NEXT STEPS

- Relational DB schema creation
- Java Model POJO classes
- DAO layer & JDBC connections
- Service layer & Auth logic
- Admin & Submission modules
- Java Swing / JavaFX Desktop UI
- Unit Testing & Final Integration

Documentation & Version Control

Repository Structure and Collaboration Pipeline

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Team & Responsibilities

Core Engineering Roles and Module Distribution



Department of Computer Science & Engineering

TKM College of Engineering (TKMCE)

B25CSB45

TEAM LEAD

MUHAMMED RINSHID VP

Team Lead

Assigned Presentation: Slides 1-4, 20

TECHNICAL MODULES:

- Project Coordination & Architecture
- System Integration & Review
- Repository & Deliverables Lead

B25CSB52

NIRMAL BINOY

Core Developer

Assigned Presentation: Slides 5-8

TECHNICAL MODULES:

- Proposed Solution Architecture
- UML Use Case Modeling
- UML Class Diagram & Package Design
- System Architecture Layering

B25CSB37

MOHAMMED NAFIH

Core Developer

Assigned Presentation: Slides 9-12

TECHNICAL MODULES:

- Role-Based Access Control (RBAC)
- Main System Decision Flowchart
- Student Submission Workflow
- Admin Resolution Workflow

B25CSB53

RAHANDEEP RD

Database & GUI Engineer

Assigned Presentation: Slides 13-14, 18

TECHNICAL MODULES:

- GUI Interface & Desktop Layouts
- Database Schema & Relational Design
- Project Timeline & Milestones Roadmap

B25CSB84

ATHIL RAHUMAN A

Documentation & Process Lead

Assigned Presentation: Slides 15-17, 19

TECHNICAL MODULES:

- Development Progress Tracking
- Documentation (PRD, TRD, Architecture)
- Team Roles & Module Mapping
- Academic References & Standards

Project Timeline

Phase Roadmap and Milestone Schedule

10-WEEK DEVELOPMENT SCHEDULE

Week-by-Week Project Timeline

Java SE • JDBC • MVC

Completed: Week 1 & 2

Current: Week 3

PROJECT PROGRESS (30% COMPLETE)

WEEK 3 OF 10

WEEK 1

Requirements & Documentation

COMPLETED

- ✓ PRD creation (prd.md)
- ✓ TRD creation (trd.md)
- ✓ Requirements & scope finalization

WEEK 2

Architecture & UML Modeling

COMPLETED

- ✓ Architecture spec (architecture.md)
- ✓ Complete UML Class Diagram
- ✓ Use Case diagram & package design

WEEK 3

Database Schema & POJO Models

IN PROGRESS

- Relational Schema definition
- Java Model POJO classes
- Enum definitions & mapping

WEEK 4

DAO & JDBC Data Access Layer

PLANNED

- JDBC Connection Factory
- UserDAO & SubmissionDAO
- SQL query optimization & mapping

WEEK 5

Service & Business Logic Layer

PLANNED

- AuthService & SubmissionService
- TrackingService
- ReportService implementation

WEEK 6

Auth & Submission Engine

PLANNED

- Password hashing & verification
- Submission creation pipeline
- Status state transition engine

WEEK 7

Admin Resolution Workflow

PLANNED

- Issue assignment engine
- Resolution note recording
- Dashboard metrics aggregation

WEEK 8

Java Desktop GUI (Swing/JavaFX)

PLANNED

- Desktop application layout
- Custom UI component styling
- Event handlers & binding

References

Academic and Technical Documentation Sources

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ACADEMIC CITATIONS

Official Standards & Specifications Consulted

[1] Oracle Corporation (2024)

Java Platform, Standard Edition (Java SE 17 / 21) API Documentation & Language Specification.

[2] Oracle JDBC Documentation

Java Database Connectivity (JDBC) API Specification for Data Access Object (DAO) Pattern.

[3] Object Management Group (OMG)

Unified Modeling Language (UML) Specification, Version 2.5.1 for Class and Use Case Diagramming.

[4] OpenJFX / JavaFX Architecture Guidelines

Desktop GUI Development Standards, Scene Builder, and Model-View-Controller (MVC) Pattern.

Closing & Q/A

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"REPORT. TRACK. RESOLVE. IMPROVE."

Thank you for your time and evaluation. We invite questions and feedback from the examination panel.

Student → AURA → University → Action